

Data Analytics Capstone

Pollution-Prevention Actions and Next-Year Toxic Release Reduction in U.S. Manufacturing

Synopsis

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Pollution-Prevention Actions and Next-Year Toxic Release Reduction in U.S. Manufacturing

Background and Context

The U.S. Environmental Protection Agency (EPA) Toxics Release Inventory (TRI) provides annual facility- and chemical-level information on toxic releases, waste management, and source-reduction activities. The TRI does not cover every chemical or every manufacturing facility, but it is a detailed, public, and reproducible source for studying reported environmental performance. Under the Pollution Prevention Act, source reduction is preferred to recycling, treatment, and disposal because it seeks to prevent waste before it is created (U.S. Environmental Protection Agency [EPA], 2025a, 2025b).

Manufacturing leaders face a practical question: do reported pollution-prevention actions correspond with a lower release of the same chemical in the following year? A simple before-and-after comparison is not enough because production changes, starting release level, chemical type, industry process, reporting differences, and unusual events can also affect annual releases. Plant managers, environmental health and safety (EHS) leaders, and process engineers need a simple way to compare records, understand which action types show stronger relationships with improvement, and identify records that may need closer review.

Past TRI research has examined public disclosure, regulatory pressure, facility behavior, and source reduction (Gamper-Rabindran, 2006; Hamilton, 1995; Konar & Cohen, 1997; Ranson et al., 2015), as well as the business case for cleaner production and profitable pollution reduction (Cagno et al., 2005; King & Lenox, 2002). Much of that work used long historical periods or causal research designs. This study has a narrower business purpose. It uses recent data, defines one practical next-year outcome, and combines statistical explanation with an interpretable prediction model. The study will estimate associations and predictive performance; it will not claim that a reported action caused a release reduction.

The analysis will use reporting years from 2020 to 2024. Action-year records from 2020, 2021, 2022, and 2023 will be matched with the same facility and chemical in the next year, creating four transitions: 2020 to 2021, 2021 to 2022, 2022 to 2023, and 2023 to 2024. The first three transitions will support model development. The 2023 to 2024 transition will remain untouched until final testing. This time-based design keeps the project realistic for six weeks and reduces the risk of using future information during model development.

Problem Statement

U.S. manufacturing facilities report many pollution-prevention actions, yet EHS and plant leaders lack a simple recent-data framework for judging whether records with those actions are more likely to achieve a meaningful reduction in the same chemical's on-site release in the following year. This project addresses that gap by examining manufacturing Form R records, measuring a meaningful next-year reduction as at least 10%, comparing reported pollution-prevention action types with records reporting no action, controlling for a small set of relevant factors in the main association model, and developing an interpretable screening model. The result is intended to support review and prioritization, not replace engineering judgment or establish causality.

Purpose of the Study

The purpose of this study is to explain and predict meaningful next-year toxic-release reductions at the facility-chemical-year level using EPA TRI Basic Plus data. The study will describe recent patterns, test the adjusted association between reporting any pollution-prevention action and next-year reduction, compare the reduction rates of individual action types with the no-reported-action group, and evaluate a regularized logistic-regression model on a future-year holdout. The business output will be a transparent screening approach that can help EHS teams focus limited review time on facility-chemical records that appear less likely to improve.

Scope and Objectives

The study population includes U.S. TRI-reporting manufacturing facilities whose primary North American Industry Classification System (NAICS) code begins with 31, 32, or 33. The unit of analysis is one facility, one TRI chemical, and one reporting year. Only Form R records will be retained because the analysis needs detailed release and source-reduction information. The core files are File 1A and File 2A for each year from 2020 through 2024. Other Basic Plus files will be excluded from the main analysis because they are not required to answer the four research questions.

The primary outcome, REDUCTION_10, equals 1 when the next-year total on-site release is no more than 90% of the current-year release and equals 0 otherwise. Percentage reduction is defined only when the current-year release exceeds zero. A missing next-year record will not be treated as a zero release. Dioxin and dioxin-like compounds reported in grams will not be combined with chemicals reported in pounds. A sensitivity analysis will exclude gram-reported chemicals.

Main Research Question

To what extent can recent EPA TRI information on reported pollution-prevention actions help explain and predict whether a U.S. manufacturing facility will reduce the same chemical's on-site release by at least 10% in the following year?

Research Questions and Hypotheses

RQ1: Do meaningful next-year reduction rates differ across action years and major manufacturing subsectors?

H0₁: The proportion of eligible records achieving REDUCTION_10 is equal across the compared years and subsectors.

H1₁: At least one compared year or subsector has a different proportion of records achieving REDUCTION_10.

RQ2: After controlling for current release level, production/activity ratio, manufacturing subsector, chemical hazard classification (EPA carcinogen and PBT status), and action year, is reporting at least one pollution-prevention action associated with achieving REDUCTION_10?

H0₂: The adjusted odds ratio for REPORTED_ANY_SOURCE_REDUCTION_ACTION equals 1.00 ($\beta_1 = 0$).

H1₂: The adjusted odds ratio for REPORTED_ANY_SOURCE_REDUCTION_ACTION differs from 1.00 ($\beta_1 \neq 0$).

RQ3: Do next-year toxic-release reduction rates differ between each reported pollution-prevention action type and records with no reported action?

H0₃: For each pollution-prevention action type, the next-year reduction rate is the same as the rate for records with no reported action.

H1₃: For at least one pollution-prevention action type, the next-year reduction rate differs from the rate for records with no reported action.

Where: α is the next-year reduction rate for action type and β is the next-year reduction rate for records with no reported action

RQ4: Can an interpretable model using only action-year information predict which facility-chemical records will fail to achieve REDUCTION_10 in the following year better than chance?

H0₄: The holdout receiver operating characteristic area under the curve equals 0.50 (AUC = 0.50).

H1₄: The holdout receiver operating characteristic area under the curve is greater than 0.50 (AUC > 0.50).

Table 1 shows how each research question contributes to the main research question and identifies the planned analytical method.

Table 1*How the Four Questions Support the Main Research Question*

Research question	Contribution to the main research question	Statistical or predictive method
RQ1	Identifies whether meaningful next-year reduction rates differ across years and manufacturing subsectors.	Descriptive proportions and confidence intervals; separate Pearson chi-square tests for year and subsector; Cramér's V effect size
RQ2	Tests whether reporting any pollution-prevention action is associated with next-year reduction after controlling for release level, production/activity, subsector, year, and chemical hazard status.	Binary logistic regression with facility-clustered standard errors
RQ3	Compares the next-year reduction rate for each reported action type with the rate for records reporting no action.	Separate 2×2 Pearson chi-square tests; Fisher's exact test when expected counts are small; Holm correction; relative risk and rate difference
RQ4	Tests whether a detailed action-year model predicts future nonreduction better than chance and better than a simple screening model.	Regularized logistic regression compared with a simple logistic-regression baseline on the 2023-to-2024 holdout

Sample Size Calculation

The study will use a separate planning calculation for each research question and will adopt the largest minimum requirement. Table 2 presents only the decision inputs and results needed in the synopsis. Appendix B provides the full concepts, formulas, variable definitions, substitutions, rounding rules, assumptions, and diagnostic limitations. Moving these technical details to an appendix keeps the main text readable while preserving a complete audit trail, consistent with the synopsis page-count rule that excludes appendices.

Table 2 summarizes the sample-size method, planning parameters, and minimum eligible-record requirement for each research question.

Table 2*Minimum Sample-Size Requirements*

Research question	Sample-size planning method	Assumed or preliminary parameter values	Minimum eligible records required
RQ1	95% confidence interval for a proportion	$p = .50$ (conservative assumption); $Z = 1.96$ (95% CI); $e = .03$ (chosen precision)	1,068
RQ2	Power approximation for a binary association	$\alpha = .05$ and power = .80 (design choices); OR = 1.30 (planning effect); action prevalence = 4.87% (preliminary data)	10,756
RQ3	Two-proportion comparison	Baseline reduction rate = .3138; detectable difference = .10; $\alpha = .01$ after conservative adjustment for five comparisons; power = .80	Approximately 290 exposed records per action type
RQ4	Prediction-model event requirement	20 parameters and 20 events/parameter (planning limits); event rate = 31.55% (preliminary data)	1,269

RQ2 produces the largest overall minimum requirement of approximately 10,756 eligible development records. The development transitions contain 123,611 eligible records and therefore exceed the overall requirement. For RQ3, adequacy must also be checked within each action type because the action groups are uneven in size. The rarest action type contains approximately 245 development records, slightly below the planning target of approximately 290 exposed records. This comparison may have limited power to detect a 10-percentage-point difference and will therefore be treated as exploratory. The observed reduction-rate difference, relative risk, confidence interval, effect size, and adjusted p-value will still be reported. The preliminary analysis contains 123,611 eligible development records and 39,967 eligible holdout records for 2023 to 2024. Therefore, the study will retain all eligible records rather than sample them. See Appendix B for the derivation of every value shown in Table 2.

Data Description

The official sources are the EPA TRI Basic Plus Data Files webpage and the related field guides (EPA, 2025a, 2025b). Broader program context on source reduction and national release trends is available through EPA resources on pollution prevention and the annual TRI National Analysis (EPA, 2025c, 2025d). File 1A supplies release, chemical, facility, unit, and industry fields. File 2A supplies production or activity information and up to four reported Source Reduction Activity Codes. The two files will be joined within each reporting year using DOCUMENT CONTROL NUMBER after applying the Form R and manufacturing filters.

Per course requirements, this project provides open access to the complete code and data pipeline through a public GitHub repository: <https://github.com/somndas3/qm640-tri-pollution-prevention-capstone>. The repository contains the README (problem statement and dataset source), a data dictionary and download instructions with a 200-row sample per reporting year, an exploratory-analysis notebook, a data-preparation script, and this synopsis. The repository folder-tree structure is provided in Appendix D.

The study will preserve the exact ZIP files and checksums used. Duplicate facility-chemical forms will be consolidated before consecutive years are matched. The detailed inclusion rules, field-level sources, code positions, formulas, numerator and denominator definitions, duplicate handling, and exclusion logic are documented in Appendix A. This keeps the main data section compact without leaving any table value undefined.

Table 3 presents the preliminary data-preparation counts for each TRI reporting year.

Table 3*Preliminary Data-Preparation Counts*

TRI reporting year	Form R manufacturing records before consolidation	Unique facility-chemical-year records after consolidation	Records reporting at least one Source Reduction Activity Code (%)
2020	57,004	56,462	4.10%
2021	56,750	56,219	5.65%
2022	56,725	56,155	5.91%
2023	55,882	55,349	5.98%
2024	54,595	54,115	5.62%

Note. Counts and rates are preliminary. The reported-action percentage equals the number of consolidated facility-chemical-year records containing at least one valid Source Reduction Activity Code, divided by all consolidated facility-chemical-year records in that year. The displayed rounded rates imply approximately 2,316 of 56,462 records in 2020, 3,174 of 56,219 in 2021, 3,319 of 56,155 in 2022, 3,311 of 55,349 in 2023, and 3,040 of 54,115 in 2024. The final validation script will report exact numerators before submission. See Appendix A, Tables A1 and A2.

Table 4 presents the preliminary counts produced by matching facility-chemical records across consecutive reporting years.

Table 4*Preliminary Consecutive-Year Matching Counts*

Action-year to outcome-year transition	Facility-chemical records present in both years	Matched records with current-year release equal to zero	Eligible matched records with current-year release greater than zero	Eligible records achieving at least a 10% next-year reduction (%)
2020 to 2021	51,907	10,605	41,302	27.55%
2021 to 2022	51,962	10,591	41,371	31.21%

Action-year to outcome-year transition	Facility-chemical records present in both years	Matched records with current-year release equal to zero	Eligible matched records with current-year release greater than zero	Eligible records achieving at least a 10% next-year reduction (%)
2022 to 2023	51,510	10,572	40,938	35.95%
2023 to 2024	50,068	10,101	39,967	32.03%

Note. A missing next-year record is excluded rather than treated as a zero release.

REDUCTION_10 is calculated only for matched records with a positive current-year release and consistent units. In the preliminary counts shown, eligible records equal matched records minus current-year zero-release records; the final validation log will separately report any exclusions for missing values, unit conflicts, or identifier conflicts. See Appendix A, Tables A3 and A4.

Core Variables

Table 5 summarizes the core analysis variables, their EPA sources or derivation rules, and their analytical roles.

Table 5

Core Analysis Variables

Analysis variable or identifier	Exact EPA source field or derivation rule	Analytical role and treatment
Facility identifier	TRIFD, Files 1A and 2A	Cross-year facility match and clustered standard errors.
Chemical identifier	TRI_CHEM_ID, Files 1A and 2A	Matches the same reported chemical across years.
Current and next-year on-site release	TOTAL ON-SITE RELEASES, File 1A	Constructs the outcome and current-release control.
Reported any source-reduction action	Derived from the first, second, third, and fourth Source Reduction Activity Code fields in File 2A	Main RQ2 predictor.

Analysis variable or identifier	Exact EPA source field or derivation rule	Analytical role and treatment
Action-type indicators	Derived by mapping valid Source Reduction Activity Codes to five prespecified action types	Binary indicators used to compare each action type with the no-reported-action group in RQ3.
Chemical hazard flags	EPA carcinogen and PBT (persistent, bioaccumulative, toxic) chemical classifications, joined by CAS NUMBER	Two binary controls (carcinogen status, PBT status) in the RQ2 model; kept to two flags to preserve model parsimony.
REDUCTION_10	Derived from current and next-year releases	Binary outcome for RQ1 to RQ4.

Appendix A contains the full data dictionary, exact EPA header labels, transformation rules, and examples. In particular, it distinguishes reported actions from unobserved actions: a value of 0 means that no valid action code was present in the selected TRI fields, not that the facility performed no environmental activity.

Analytic Approach

The main text summarizes the method chosen for each question. Appendix C provides the full examiner-safe structure for every question: concept, mathematical principle, variable definitions, setup, decision rule, business interpretation, and diagnostic limitations. Placing symbolic specifications and metric formulas in Appendix C avoids repeating technical detail in the main narrative.

Table 6 summarizes the statistical or predictive method and hypothesis decision rule for each research question.

Table 6

Analysis Plan by Research Question

Research question	Statistical or predictive method	Reported result and hypothesis decision rule
RQ1	Chi-square test. Two separate tests (year by outcome; subsector by	Report differences in REDUCTION_10 rates; reject H_0 when $p < .05$, while emphasizing

Research question	Statistical or predictive method	Reported result and hypothesis decision rule
	outcome), not a joint interaction model. Proportions, confidence intervals, Pearson chi-square test, and Cramer's V reported for each.	effect size.
RQ2	Logistic regression. Binary logistic regression with facility-clustered standard errors.	Report the adjusted odds ratio and 95% confidence interval for REPORTED_ANY_SOURCE_REDUCTION_ACTION; reject H_{02} when the interval excludes 1.00 and $p < .05$.
RQ3	Chi-square test / Fisher's exact test. Separate 2×2 comparisons of each action type with the no-reported-action group. Pearson chi-square will be used when expected cell counts are adequate; Fisher's exact test will be used when counts are small. Holm-adjusted p-values will address multiple comparisons.	Report group counts, reduction rates, percentage-point differences, relative risks, 95% confidence intervals, effect sizes, and Holm-adjusted p-values. Reject an action-type-specific null hypothesis when the Holm-adjusted p-value is below .05.
RQ4	Logistic regression (regularized/penalized), compared against a simple unregularized baseline (reported-action indicator plus current release only). Both trained on 2020 to 2022 action years and evaluated once, on the same 2023 to 2024 holdout.	Test the main model's AUC against .50 and against the baseline model's AUC; assess Brier score, calibration, precision, recall, and F1 for both models on the same holdout.

Action types are not mutually exclusive. A record reporting more than one action type may appear in more than one separate comparison, but the reference group will always contain records reporting no valid action. RQ4 will use only information available in the action year. The elastic-net penalty strength and mixing parameter will be selected by k-fold cross-validation within the development transitions only, using folds grouped by reporting year so no fold mixes training and

validation records from the same transition; the model will then be locked and evaluated once on the 2023 to 2024 holdout. No performance value will be promised before testing.

Sensitivity and Quality Checks

The RQ2 association model and the RQ3 group comparisons will be repeated without the 2020-to-2021 transition (which spans a pandemic-disrupted production period that could distort normal release patterns) and without gram-reported chemicals. The analysis will also document missingness, unit conflicts, extreme production or activity ratios, sparse action-type counts, small expected cell counts, logistic-model separation, multicollinearity, influential records, calibration, and subgroup performance. Appendix C states the diagnostic rule and response for each issue.

Three limitations of the outcome and exposure variables are stated directly rather than left implicit. First, REDUCTION_10 cannot distinguish a genuine pollution-prevention-driven reduction from a release drop caused by a production slowdown, a facility closure, or the discontinued use of that chemical; TRI fields alone cannot separate these cases, so results should be read as screening associations, not confirmed causal improvements. Second, a facility-chemical record whose release falls far enough to drop below the EPA reporting threshold will stop filing and will appear as a missing next-year record, which this study excludes rather than scores as REDUCTION_10 = 1; this likely removes some of the strongest true reductions from the analysis and biases estimated reduction rates and associations conservatively (downward), which is a limitation of the design rather than an error. Third, REPORTED_ANY_SOURCE_REDUCTION_ACTION depends on facility self-reporting of Source Reduction Activity Codes, and a value of 0 does not prove that no action was taken; if this underreporting is unrelated to the outcome, it will attenuate the RQ2 and RQ3 associations toward the null, meaning the true association could be stronger than what is observed. Finally, the RQ2 model's release-level control addresses, but should be read with, ordinary regression-to-the-mean: a facility-chemical record with an

unusually high current-year release is mechanically more likely to clear a 10-percentage-point relative bar the following year regardless of any reported action.

The scope remains limited to two source files per year, one primary outcome, four linked research questions, one main prediction model, and one comparison model (a simple, unregularized logistic regression using only the reported-action indicator and current release, evaluated on the same holdout to show whether the main model's added predictors and regularization earn their added complexity). Detailed definitions are written once in the appendices and cited from the main text, reducing repetition and helping the core synopsis remain within the required page range.

Recommendation and Application

The target users are plant managers, EHS leaders, process engineers, corporate environmental managers, and TRI analysts. The output can help them compare the observed next-year reduction rates associated with different reported action types and prioritize facility-chemical records that appear unlikely to achieve a meaningful next-year reduction. It will not prescribe an engineering change or establish noncompliance.

Recommendations will depend on the observed adjusted associations, uncertainty, and holdout performance. The site EHS leader is the proposed process owner, supported by plant management and process engineering. Human technical review will remain the final decision point.

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Appendix A

Data Definitions, Derivations, and Table Construction Rules

This appendix defines every displayed data count and derived variable. It is referenced in the Data Description section so that the main synopsis remains concise while the calculation trail remains complete and reproducible. Exact EPA field labels and column positions are verified against the TRI Basic Plus field guide (EPA, 2025b).

Table A1 defines the population filters, record units, consolidation rules, and rate calculations used to construct the preliminary data-preparation counts.

Table A1

Definitions for the Preliminary Data-Preparation Counts

Table column heading	Operational definition	Exact derivation or counting rule
TRI reporting year	The REPORTING YEAR value in the national TRI Basic Plus files.	One table row is produced for each year from 2020 through 2024.
Form R manufacturing records before consolidation	Rows that remain after retaining FORM TYPE = R and a PRIMARY NAICS CODE that begins with 31, 32, or 33.	Count raw qualifying File 1A records after the one-to-one File 1A/File 2A join by DOCUMENT CONTROL NUMBER and before duplicate facility-chemical forms are consolidated.
Unique facility-chemical-year records after consolidation	Distinct combinations of TRIFD, CAS NUMBER, and CHEMICAL NAME within a reporting year, after resolving duplicate Form R filings for the same facility-chemical-year with a deterministic tie-break rule.	Group by TRIFD + CAS NUMBER + CHEMICAL NAME within REPORTING YEAR. Tie-break rule for duplicate filings: REPORTED_ANY_SOURCE_REDUCTION_ACTION = 1 if any duplicate filing reports an action (logical OR across duplicates); the authoritative TOTAL ON-SITE RELEASES and UNIT OF MEASURE are taken from the filing with the highest DOCUMENT CONTROL NUMBER among the duplicates. TRI_CHEM_ID is retained on the consolidated record for cross-year

Table column heading	Operational definition	Exact derivation or counting rule
		matching (Table A2).
Records reporting at least one Source Reduction Activity Code (%)	Percentage of consolidated facility-chemical-year records with at least one valid reported action code in the four File 2A activity-code fields.	Numerator = count(REPORTED_ANY_SOURCE_REDUCTION_ACTION = 1). Denominator = all consolidated facility-chemical-year records for the year. Rate = numerator / denominator × 100.

Table A2 explains how the four EPA Source Reduction Activity Code fields are converted into the reported-action indicator and the five action-type indicators.

Table A2

Derivation and Coding of Reported Source-Reduction Variables

Analysis variable	Exact EPA source fields or derivation rule	Coding or calculation rule
Source Reduction Activity Code	File 2A columns 145, 155, 165, and 175: FIRST, SECOND, THIRD, and FOURTH SOURCE REDUCTION ACTIVITY CODE.	Each nonblank valid code describes an action reported for the chemical on that Form R.
REPORTED_ANY_SOURCE_REDUCTION_ACTION	Derived from the four activity-code fields after duplicate forms are consolidated.	1 if at least one valid code is present; 0 if all four consolidated fields are blank or contain no valid code.
Action-type indicators	Valid codes mapped to material modification, product modification, process/equipment modification, inventory/material management, and operating practice/training.	Each action type is coded 1 when one or more mapped codes are present and 0 otherwise. Action types may co-occur. Legacy W codes are retained separately until their treatment is justified.

$\text{REPORTED_ANY_SOURCE_REDUCTION_ACTION}_i = I(\text{CODE1}_i \vee \text{CODE2}_i \vee \text{CODE3}_i \vee \text{CODE4}_i \text{ is valid})$

The five action-type indicators are not mutually exclusive. A facility-chemical-year record may be coded 1 for more than one action type when multiple valid Source Reduction Activity Codes are reported. For RQ3, each action type is compared separately with records reporting no valid source-reduction action. Here, $I(.)$ is an indicator function. A value of 1 confirms that at least one action code was reported in the selected File 2A fields. A value of 0 does not prove that no pollution-prevention activity occurred; it means only that no valid code was observed in those fields.

Table A3 defines every column used in the consecutive-year matching table and states the exact counting and exclusion rules.

Table A3

Definitions for Consecutive-Year Matching Counts

Table column heading	Operational definition	Exact derivation or counting rule
Action-year to outcome-year transition	A pair of consecutive reporting years used to connect predictors to a later outcome.	2020 to 2021, 2021 to 2022, 2022 to 2023, and 2023 to 2024.
Facility-chemical records present in both years	Consolidated action-year records matched to the same TRIFD and TRI_CHEM_ID in the following year.	Inner join action year t to year $t+1$ using TRIFD + TRI_CHEM_ID; validate CAS number and unit consistency.
Matched records with current-year release equal to zero	Matched records for which current-year TOTAL ON-SITE RELEASES = 0.	Count matched records with $R_t = 0$. These are excluded from percentage reduction because the denominator would be zero.
Eligible matched records with current-year release greater than zero	Matched records with $R_t > 0$, a nonmissing R_{t+1} , and consistent units.	Eligible N = matched records - current-year zero records - records failing missingness or unit checks.
Eligible records achieving	Percentage of eligible matched	Numerator = count(REDUCTION_10 =

Table column heading	Operational definition	Exact derivation or counting rule
at least a 10% next-year reduction (%)	records for which the next-year release is no more than 90% of the current-year release.	1). Denominator = eligible matched records. Rate = numerator / denominator $\times 100$.

Table A4 presents the mathematical derivation of the primary outcome variable, REDUCTION_10.

Table A4

Primary Outcome Derivation : $\text{REDUCTION_10}_i = 1$ when $R_{i,t+1} \leq 0.90 \times R_{i,t}$; otherwise

$\text{REDUCTION_10}_i = 0$, provided $R_{i,t} > 0$. Percentage change_i = $[(R_{i,t+1} - R_{i,t}) / R_{i,t}] \times 100$

$R_{i,t}$ is the total on-site release for facility-chemical record i in the action year, and $R_{i,t+1}$ is the same facility and chemical in the following year. A next-year zero is a valid reduction. A missing next-year record is not converted to zero. Pound and gram quantities are never added together.

Table A5 provides the field-level data dictionary and analytical treatment of each core variable.

Table A5

Core Field Dictionary and Transformation Notes

EPA field or derived analysis variable	EPA source file or derivation	Operational definition and analytical treatment
DOCUMENT CONTROL NUMBER	Files 1A and 2A	Within-year form-level join key. Uniqueness and join cardinality are audited before the analysis.
TRIFD	Files 1A and 2A	Facility identifier used for cross-year matching and facility-clustered standard errors.
TRI_CHEM_ID	Files 1A and 2A	Primary chemical identifier for consolidation and cross-year matching. CAS NUMBER is a validation field.
PRIMARY NAICS CODE	Files 1A and 2A	Filter to codes beginning with 31, 32, or 33; first three digits define the manufacturing subsector.
TOTAL ON-SITE RELEASES	File 1A	Reported annual quantity used for the current-release control and next-year outcome. $\log(1 + \text{release})$ is used as

EPA field or derived analysis variable	EPA source file or derivation	Operational definition and analytical treatment
PROD RATIO/ACTIVITY INDEX	File 2A	a control when appropriate. Current-year production or activity control. Duplicate values are summarized by the median; missingness and extreme values are reported.
UNIT OF MEASURE	File 1A	Used to prevent an invalid combination of pounds and grams. Inconsistent matched pairs are excluded from percentage-change calculations.
Chemical hazard classification (carcinogen, PBT)	EPA TRI chemical reference list, matched by CAS NUMBER (EPA, 2025b)	Two binary RQ2 model controls: 1 if the chemical is EPA-classified as a carcinogen for TRI reporting-threshold purposes; 1 if it is EPA-classified as a persistent, bioaccumulative, and toxic (PBT) chemical. Limited to two flags, not a full chemical-identity control, to avoid inflating the parameter count.

Appendix B

Sample-Size Calculations and Substitution Details

This appendix contains the full calculations supporting Table 2. Each substituted value is identified as a statistical constant, a researcher-selected planning assumption, a published methodological planning rule, or a preliminary rate calculated from the development transitions. Preliminary rates are feasibility inputs rather than final findings.

RQ1: Confidence-Interval Precision

Concept: Estimate a proportion with a 95% confidence interval and a margin of error of .03.

Formula or Mathematical Principle: $n = Z^2 p(1 - p) / e^2$.

$$n = (1.96^2 \times .50 \times .50) / .03^2 = 1,067.11$$

Variable Definition: n is the required number of eligible records; Z = 1.96 is the two-sided 95% standard-normal critical value; p = .50 is the conservative assumed proportion; e = .03 is the selected margin of error.

Substitution, Setup, or Step-by-Step Explanation: Substitute Z = 1.96, p = .50, and e = .03 into the formula: $n = (1.96^2 \times .50 \times .50) / (.03^2) = 1,067.11$. Because a partial record cannot be used, round upward.

Final Answer: Round upward to 1,068 eligible records.

Limitation, Risk, or Diagnostic Note: Precision is lower within small subgroups, even when the overall N is adequate.

RQ2: Binary-Association Power Approximation

Concept: Plan for a small association when reported action exposure is uncommon.

$$p_1 = (OR \times p_0) / [1 - p_0 + (OR \times p_0)]$$

$$p_1 = (1.30 \times .3138) / [1 - .3138 + (1.30 \times .3138)] = .37284$$

Variable Definition: $p_0 = .3138$ is the preliminary REDUCTION_10 probability among eligible development records without a reported action; p_1 is the expected probability among records with a reported action; $OR = 1.30$; $q = .0487$ is the exposed-group allocation; $r = (1 - q)/q$ is the unexposed-to-exposed allocation ratio; $\alpha = .05$; power = .80; $Z_{1-\alpha/2} = 1.96$; and $Z_{1-\beta} = 0.842$.

Formula and Substitution: Let $r = (1 - .0487)/.0487 = 19.534$. Let $\bar{p} = (p_1 + rp_0)/(1 + r) = .31668$. The required exposed-group size is $n_1 = \{Z_{1-\alpha/2}\sqrt{[(1 + 1/r)\bar{p}(1 - \bar{p})] + Z_{1-\beta}\sqrt{[p_1(1 - p_1) + p_0(1 - p_0)/r]}}\}^2/(p_1 - p_0)^2$. Substitution gives $n_1 = 523.79$. Therefore, total $n = n_1(1 + r) = 10,755.51$, rounded upward.

Final Answer: RQ2 requires approximately 10,756 eligible development records under the stated planning assumptions.

Limitation, Risk, or Diagnostic Note: This is a transparent planning approximation. Facility clustering can increase uncertainty in the fitted model.

RQ3: Two-Proportion Comparison

Concept: Compare the next-year reduction rate for one action type with the rate for records reporting no action.

Formula: Where: p_0 , the preliminary no-action reduction rate, p_1 , representing a 10-percentage-point difference, α , a conservative level for five planned comparisons, power, r , is the ratio of no-action records to action-type records. For a very large no-action reference group, the required exposed action-type sample is approximately 290 records.

Final answer: Approximately 290 records are required in each action-type group to detect a 10-percentage-point difference under the stated assumptions.

Limitation: The rarest action type has approximately 245 development records and may not meet this planning target. That comparison will be labelled exploratory and evaluated with Fisher's exact test and confidence intervals.

The rarest action type contains approximately 245 development records, slightly below the planning target. That comparison may be underpowered for detecting a 10-percentage-point difference and will therefore be treated as exploratory. Fisher's exact test, confidence intervals, effect size, and the observed rate difference will still be reported.

RQ4: Prediction-Model Planning

$$\text{Required events} = 20 \times P = 20 \times 20 = 400$$

$$n = \text{required events} / \text{event rate} = 400 / 0.3155 = 1,268.03$$

Variable Definition: P = the prespecified maximum of 20 effective predictor parameters; 20 is the selected events-per-parameter planning screen; event rate = .3155 is the preliminary development REDUCTION_10 rate; n is the required development records.

Substitution, Setup, or Step-by-Step Explanation: First calculate required events: 20 events per parameter $\times$ 20 parameters = 400 events. Then divide by the preliminary event rate: $n = 400 / .3155 = 1,268.03$. Round upward.

Final Answer: Round upward to 1,269 development records, plus a separate holdout set.

Limitation, Risk, or Diagnostic Note: This is a planning screen based on the events-per-variable heuristic (Peduzzi et al., 1996) rather than the final Riley-style calculation. Calibration, shrinkage, predictor prevalence, and overfitting will also be evaluated.

Appendix C

Research-Question Models, Decision Rules, and Evaluation Metrics

RQ1 Mathematical Specification

$$\hat{p}_g = y_g/n_g \quad , \quad \chi^2 = \sum_i \sum_j [(O_{ij} - E_{ij})^2 / E_{ij}]$$

y_g is the number of REDUCTION_10 events in group g ; n_g is the eligible count; O_{ij} and E_{ij} are observed and expected contingency-table counts. Year and subsector are each tested against REDUCTION_10 in a separate two-way contingency table (year by outcome, then subsector by outcome), not as a single joint interaction model, to keep the test simple and directly interpretable. Report confidence intervals and Cramer's V for each table because a large N can make minor differences statistically significant.

RQ2 Mathematical Specification

$$\log[p_i/(1 - p_i)] = \beta_0 + \beta_1 A_i + \beta_2 \log(1 + R_i) + \beta_3 P_i + \gamma^T X_i$$

A_i is REPORTED_ANY_SOURCE_REDUCTION_ACTION; R_i is the current-year on-site release; P_i is the production or activity ratio; X_i contains the year, three-digit NAICS subsector, and two chemical hazard flags: EPA-classified carcinogen status and EPA-classified persistent, bioaccumulative, and toxic (PBT) status (see Table A5). $\exp(\beta_1)$ is the adjusted odds ratio. Reject H_{02} when its 95% confidence interval excludes 1.00 and $p < .05$.

RQ3 Mathematical Specification

For each action type :

$$, , ,$$

Where: $n_{g,t}$ is the number of REDUCTION_10 events among records reporting action type t , $n_{g,0}$ is the number of eligible records reporting action type t , $n_{g,1}$ is the number of REDUCTION_10 events among records reporting no action, $n_{g,2}$ is the number of eligible records reporting no action, Δ is the percentage-point difference in reduction rates, RR is the relative risk, and O_{ij} and E_{ij} are the observed and expected counts

A separate 2×2 table is created per action type, using Pearson chi-square when expected counts are adequate and Fisher's exact test when they are small, with Holm-adjusted p-values across the five comparisons. Action types are nonexclusive, so a record may appear in more than one comparison; the no-action reference group stays the same throughout.

RQ4 Penalized Prediction Model

$$L(\beta) = -\sum_i [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + \lambda[(1 - \alpha_r) \|\beta\|_2^2 / 2 + \alpha_r \|\beta\|_1]$$

λ controls penalty strength, and α_r controls the elastic-net mix. All preprocessing and tuning occur within the development period. The 2023 to 2024 transition is used once as the final holdout.

Comparison model: a simple, unregularized logistic regression fit on the same development records, using only REPORTED_ANY_SOURCE_REDUCTION_ACTION and $\log(1 + \text{current-year release})$ as predictors: $\log[p_i / (1 - p_i)] = \gamma_0 + \gamma_1 A_i + \gamma_2 \log(1 + R_i)$. It requires no tuning and is evaluated once on the same 2023-to-2024 holdout with the same metrics as the main penalized model (Table C1). Reporting both side by side shows whether the main model's added predictors and regularization deliver real predictive value beyond this screening rule.

Table C1 defines the prediction metrics used to assess final holdout performance of both models, following established guidance on prediction-model performance and calibration (Steyerberg et al., 2010; Van Calster et al., 2019).

Table C1

Prediction Metrics and Interpretation

Metric	Formula or definition	Interpretation
Precision	$TP / (TP + FP)$	Among records flagged as positive, the proportion was correctly flagged.
Recall	$TP / (TP + FN)$	Among all actual positive records, the proportion is detected.
F1	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$	Balances precision and recall for the

Metric	Formula or definition	Interpretation
		stated positive class.
Brier score	$(1/n) \sum (p_i - y_i)^2$	Mean squared probability error; lower is better.
ROC-AUC	Probability that a randomly selected positive record receives a higher score than a randomly selected negative record.	Ranking performance across thresholds; .50 represents chance ranking.
Calibration	Agreement between predicted probabilities and observed event rates.	Assessed with a calibration plot and calibration summaries.

For operational reporting, nonreduction may be coded as the positive class. The class definition will be stated before metrics are calculated and will not change across analyses.

Table C2 presents the prespecified diagnostic checks and the planned response when each issue is identified.

Table C2*Prespecified Diagnostic Checks*

Risk or diagnostic	Check	Planned response
Missing data	Report missingness by variable and year.	Use complete cases when missingness is small; otherwise use a simple development-only imputation or missing category and disclose it.
Unit conflict	Compare UNIT OF MEASURE across each matched pair.	Exclude inconsistent pairs from percentage-change calculations; report excluded counts.
Sparse action type	Review action-type counts, outcome counts, and expected cell frequencies.	Use Fisher's exact test when needed; report confidence intervals; label underpowered comparisons exploratory.
Logistic-model multicollinearity	Inspect correlation and variance-inflation diagnostics for RQ2 and RQ4 predictors.	Remove or regroup redundant model predictors.
Multiple comparisons in RQ3	Review the five planned action-type p-values	Apply Holm adjustment and interpret adjusted p-values together with effect sizes
Separation or unstable logistic estimates	Inspect convergence, large coefficients, and wide intervals. Applies to RQ2 and RQ4 only.	Apply regularization or simplify the model; do not present unstable rankings.
Temporal disruption	Repeat association models without 2020 to 2021.	Compare direction and magnitude with the primary result.
Prediction drift or poor calibration	Compare holdout discrimination and calibration with development results.	Do not recommend operational screening when holdout performance is weak.

Appendix D

GitHub Repository and Folder Structure

Per course requirements, the complete code and data pipeline for this project are maintained in a public GitHub repository with open access for reviewers:

<https://github.com/somndas3/qm640-tri-pollution-prevention-capstone>

The repository contains the README, a data folder with EPA download instructions and a 200-row sample per reporting year, an exploratory-analysis notebook, a data-preparation script, and this synopsis document.

qm640-tri-pollution-prevention-capstone/

```
|-- README.md          (problem statement + dataset source)

|-- requirements.txt

|-- data/

|   |-- README.md      (EPA download links + reproduction instructions)
|   |-- sample/        (200-row sample, File 1A + 2A, 2020-2024)
|-- notebooks/

|   |-- 01_data_exploration.ipynb (shape, head, dtypes, missing values, Table 3 reconciliation)

|-- src/

|   |-- data_prep.py    (Form R + manufacturing NAICS filter + consolidation, 1A/2A join)

|-- reports/

    |-- QM640_Synopsis_TRI_2020_2024.docx
```